

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Control and Coordination

CBSE Science 2026-27 | Board-examinable

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

What is a reflex action, and how does a reflex arc help?

Answer:

A reflex is a rapid, automatic response to a stimulus. A reflex arc carries a signal from receptor through sensory neuron and spinal cord to a motor neuron, enabling quick action.

Question 2

How does a shoot bend towards light?

Answer:

Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Unequal growth bends the shoot towards the light.

Question 3

Give one difference between nervous and hormonal coordination.

Answer:

Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons. Hormones travel through blood and generally act more slowly but may have longer-lasting effects.

Question 4

What is the central idea or skill tested in Control and Coordination?

Answer:

The central focus is: A reflex is a rapid, automatic response to a stimulus. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on Control and Coordination.

Answer:

First point: A reflex is a rapid, automatic response to a stimulus. Second point: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in Control and Coordination lead to an important result or conclusion?

Answer:

Begin with the key idea: A reflex is a rapid, automatic response to a stimulus. Then connect it to the result: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there.

Control and Coordination - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in Control and Coordination?

Answer:

Use this supporting detail: A reflex is a rapid, automatic response to a stimulus. A reflex arc carries a signal from receptor through sensory neuron and spinal cord to a motor neuron, enabling quick action. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in Control and Coordination.

Answer:

Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Unequal growth bends the shoot towards the light. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from Control and Coordination?

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons.

Question 10

Compare the first and second important ideas in Control and Coordination.

Answer:

The first idea is: A reflex is a rapid, automatic response to a stimulus. The second idea is: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from Control and Coordination.

Answer:

Cause or starting point: A reflex is a rapid, automatic response to a stimulus. Effect or outcome: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of Control and Coordination?

Answer:

A clear sequence is: A reflex is a rapid, automatic response to a stimulus. Next, Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Finally, Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons.

Control and Coordination - Continued

Question 13

Give a 40-50 word answer outline for an important question from Control and Coordination.

Answer:

Start with the direct answer: A reflex is a rapid, automatic response to a stimulus. Add one supporting point: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. End with the outcome or message: Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in Control and Coordination?

Answer:

Use an accurate scientific statement: A reflex is a rapid, automatic response to a stimulus. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, Control and Coordination?

Answer:

Connect the idea to its reason or result. For instance: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in Control and Coordination.

Answer:

A sound inference is that nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in Control and Coordination.

Answer:

Apply the concept by first stating: A reflex is a rapid, automatic response to a stimulus. Then justify the response with: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there.

Question 18

What would make an answer from Control and Coordination incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: A reflex is a rapid, automatic response to a stimulus. Also add: Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons.

Control and Coordination - Continued

Question 19

Prepare an Assertion-Reason style explanation based on Control and Coordination.

Answer:

Assertion: A reflex is a rapid, automatic response to a stimulus. Reason: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from Control and Coordination in two clear steps?

Answer:

Step 1: state the concept or event - A reflex is a rapid, automatic response to a stimulus. Step 2: support it with the explanation or result - Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there.

Question 21

What answer structure is suitable for a 3-mark question from Control and Coordination?

Answer:

Use three connected points: A reflex is a rapid, automatic response to a stimulus. Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from Control and Coordination?

Answer:

Give an introduction, develop these key points - A reflex is a rapid, automatic response to a stimulus. Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. - and close with: Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons. Add a diagram, equation, example, or textual reference where relevant.

Question 23

How can the ideas in Control and Coordination be applied to a new situation or case study?

Answer:

Identify the relevant principle first: A reflex is a rapid, automatic response to a stimulus. Then use the chapter's reasoning to explain the case: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there.

Question 24

Which detail from Control and Coordination is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons. Explain it in context instead of using it as an isolated statement.

Control and Coordination - Continued

Question 25

Write a brief concluding statement for an answer on Control and Coordination.

Answer:

A suitable conclusion is: Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons. This conclusion should follow naturally from the explanation rather than introduce a new point.

Question 26

Make a three-point revision note for Control and Coordination.

Answer:

1. A reflex is a rapid, automatic response to a stimulus. 2. Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. 3. Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from Control and Coordination before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - A reflex is a rapid, automatic response to a stimulus. - and supported it with a reason, example, calculation, or outcome such as: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there.

Question 28

Why does Control and Coordination remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: A reflex is a rapid, automatic response to a stimulus. Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons.

Question 29

Create a final 60-80 word answer plan for a long question from Control and Coordination.

Answer:

Opening: A reflex is a rapid, automatic response to a stimulus. Development: Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there. Conclusion: Nervous coordination uses electrical impulses for fast, usually short-lived responses along neurons. Use complete sentences and include all required evidence, steps, or references.

Question 30

What is one last-minute revision question you should ask yourself about Control and Coordination?

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: A reflex is a rapid, automatic response to a stimulus. and Auxin gathers more on the shaded side of the shoot and promotes greater cell elongation there.

Sources: CBSE Science curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.